Introduction to Statistical Investigations

Second Edition

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Chapter 3.1

Estimation
Statistical Inference: Confidence Intervals

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Chapter 3 Overview

- So far, we can only say things like
  - “We have strong evidence that the probability Harley chooses the correct cup is > 0.5.”
  - ”Students overestimate the length of the Jackson 5’s song ‘ABC.’”
- We want a method that estimates what the parameter *actually is.*

Can Dogs Sniff Out Cancer?

Marine, a dog originally trained for water rescues, was tested to see if she could detect if a patient had colorectal cancer by smelling a sample of the patient's breath.

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Can Dogs Sniff Out Cancer?

- In each trial, she first smells a bag from a patient with colorectal cancer.
- Then she smells 5 other samples; 4 from normal patients and 1 from a person with colorectal cancer.
- Marine correctly identified the bag from the cancer patient in 30 out of 33 trials. So our sample proportion is 30/33 = 0.909.

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Can Dogs Sniff Out Cancer?

- In each of the 33 trials, Marine smells 5 bags containing breath samples: 4 from normal patients and 1 from a person with colorectal cancer.

- Discuss:
  - What are the observational units?
  - What is the variable?
  - What is our sample size?
  - What's the general probability of success, π?
  - Write the null and alternative hypotheses in symbols.

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Can Dogs Sniff Out Cancer?

Our sample proportion lies more than 10 standard deviations above the mean and hence our p-value is 0.

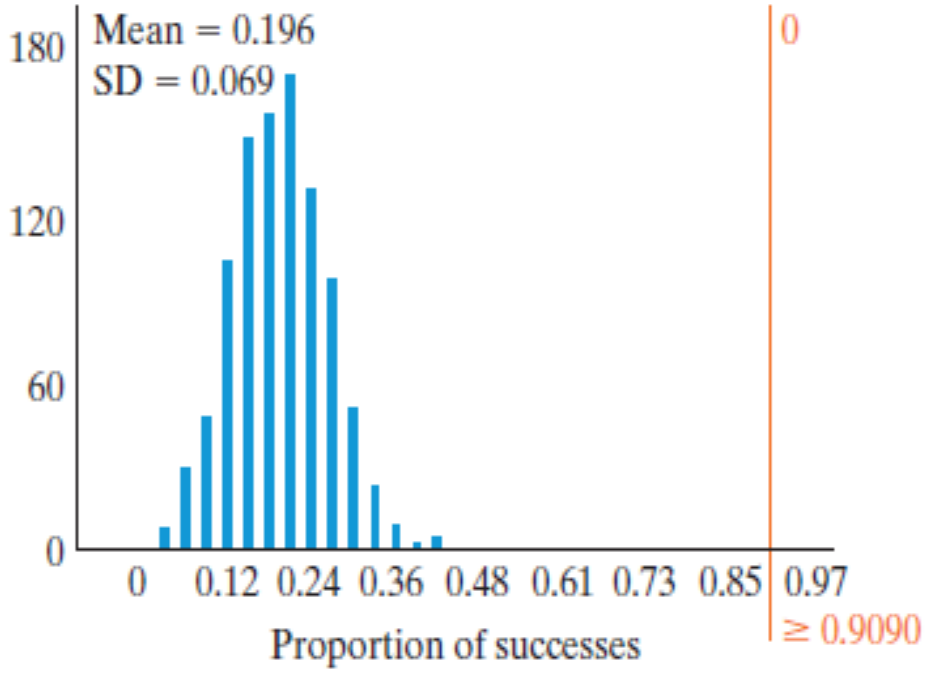
Probability of success (π): 0.20

Sample size (n): 33

Number of samples: 1000

As extreme as ≥ 0.909 Count

Proportion of samples:
0/1000 = 0



Can Dogs Sniff Out Cancer?

- It's not very interesting to say "Marine picks the correct bag more than 20% of the time."
- Can we say for certain that Marine picks the correct bag 90.9% of the time?

- Is it plausible that Marine actually picks the correct bag 70% of the time and just had a lucky day?

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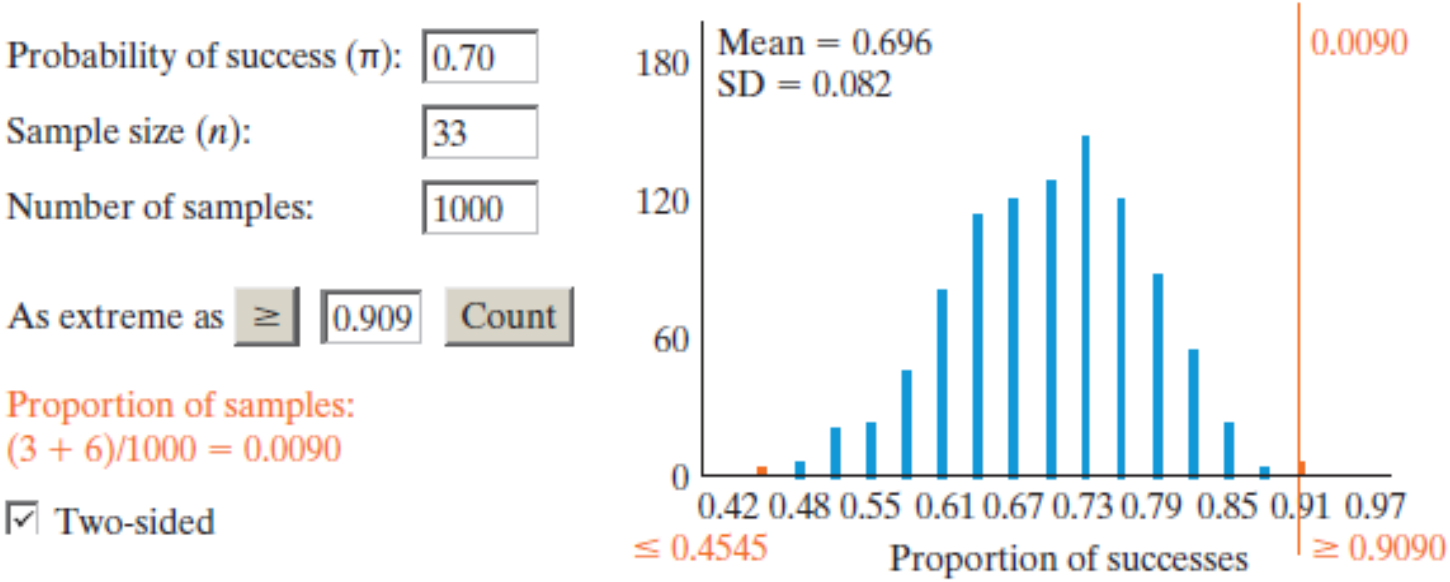
Can Dogs Sniff Out Cancer?

- Let's try:
- H_0: $\pi = 0.70$ H_a: $\pi \neq 0.70$
- We get a small p-value (0.0090) so we can rule out 0.70 as a value for π.



Can Dogs Sniff Out Cancer?

- What about 0.80?
- Is an observed statistic of 0.909 unlikely if $\pi = 0.80$?

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Can Dogs Sniff Out Cancer?

- H_0: $\pi = 0.80$ H_a: $\pi \neq 0.80$
- We get a large p-value (0.1470) so 0.80 is a ***plausible*** value for Marine's long-run probability.

Probability of success (π): 0.80

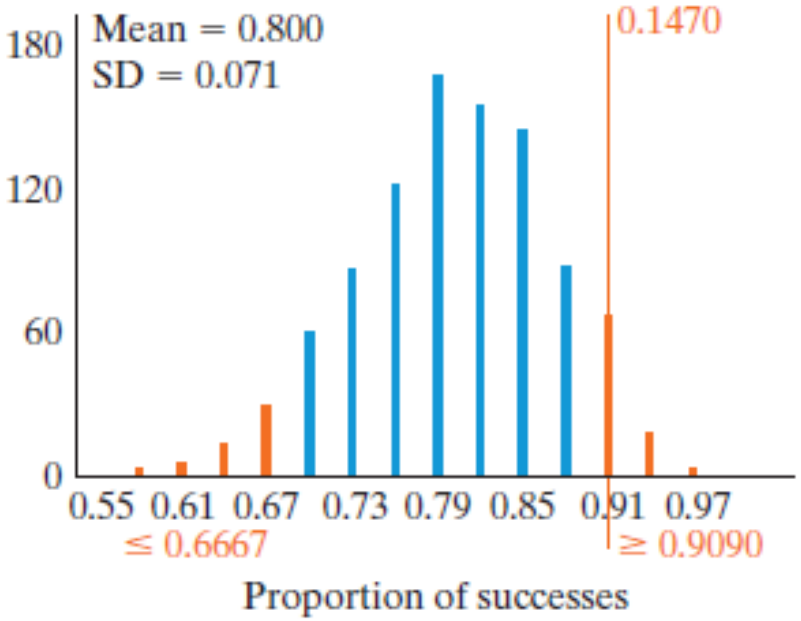
Sample size (n): 33

Number of samples: 1000

As extreme as ≥ 0.909 Count

Proportion of samples:
(52 + 95)/1000 = 0.1470

☑ Two-sided



A range of plausible values

- Small p-value => the value in H_0 is not plausible. (We can conclude H_a.)
- Large p-value => the value under H_0 is plausible. (We can't conclude H_a.)
- The **significance level** α determines whether a p-value is small or large.

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A range of possible values

- We can use the One Proportion applet to find a range of plausible values for Marine's long-term probability of choosing the correct bag.
- We do this by keeping the sample proportion the same and varying π.
- You will do this in today's Exploration.

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Exploration 3.1